Intel® Pentium® 4 Processor and Intel® 850 Chipset Family Performance Brief

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EXECUTIVE SUMMARY: THE INTEL® PENTIUM® 4 PROCESSOR AND INTEL® 850 CHIPSET FAMILY

The Intel® Pentium® 4 processor is Intel's most advanced and powerful processor for desktop PCs, offering several innovative features for maximum performance, productivity, and manageability, and is based on the new Intel NetBurst™ micro-architecture. The Pentium 4 processor is designed to deliver performance across applications and usage models where PC users can truly appreciate and experience performance. For example, these applications include Internet audio and video streaming, digital image processing, digital video content creation, speech recognition, 3D graphics, CAD, multimedia, and multi-tasking user environments. The Intel Pentium 4 processor delivers world-class performance for consumer enthusiasts, business professionals, as well as for entry-level workstation users. With all the performance needed to maximize both current and future Web technologies, as well as being designed to maximize investment protection, Pentium 4 processors will continue to deliver world-class performance for PC users well into the future.

The Pentium 4 processor at 1.50 GHz, 1.60 GHz, 1.70 GHz, 1.80 GHz, 1.90 GHz, 2 GHz, 2A GHz, 2.20 GHz, 2.26 GHz, 2.40 GHz, 2.53 GHz, 2.66 GHz and 2.80 GHz enables high performance through the innovative NetBurst micro-architecture—including Hyper Pipelined Technology, 533 MHz and 400 MHz System Bus speed (complemented by the corresponding bandwidth enabled by the 850E and 850 chipset respectively), Execution Trace Cache, and Rapid Execution Engine—in addition to a number of enhanced features—including the Advanced Transfer Cache, Advanced Dynamic Execution, Enhanced Floating-point and Multimedia units, and Streaming SIMD Extensions 2 (SSE2)—which provide a powerful combination to meet the higher performance and data bandwidth needs for today's and tomorrow's computing environment.

The Intel Pentium 4 processor provides awesome performance for the Visual Internet and is fully compatible with existing Intel Architecture-based software. The Pentium 4 processor takes desktop performance into the future by offering performance headroom designed to maximize the life of your PC investment. Software designed for the Pentium 4 processor unleashes its full multimedia capabilities, including full-screen and full-motion video, realistic graphics, and an enhanced, exciting Internet experience.

When a Pentium 4 processor is used in conjunction with an Intel® 850 chipset family, the resulting platform provides the highest level of performance for today's software and performance headroom for tomorrow's emerging software. With smoother audio and video, faster and more realistic 3D, and unmatched responsiveness for Internet applications, the Pentium 4 processor combined with the Intel 850 chipset family provides best-in-class performance for today's and tomorrow's performance desktops.

The Intel 850 chipset family implements the next step in the evolution of the Intel Hub Architecture and was designed in tandem with the Intel Pentium 4 processor and innovative Intel NetBurst™ micro-architecture. The Intel 850 chipset family was designed to maximize performance and enhance the PC user experience both on and off the Internet. Features such as a 533 MHz and 400 MHz system bus offer a peak bandwidth of 4.2 GB/s and 3.2 GB/s, Dual RDRAM memory channels also offer a peak bandwidth of 4.2 GB/s or 3.2 GB/sec data transfer rate that is balanced with the respective system bus, and the AGP 4X interface allows graphics controllers to access main memory at over 1 GB/s. Intel Hub Architecture delivers twice the I/O bandwidth as previous generation northbridge/southbridge technology. With dedicated data paths to fully optimize the additional bandwidth, the Intel 850 chipset family offers significant gains in the Performance PC market and support for future Intel NetBurst micro-architecture-based processors.

Modern desktop systems are used to run a broad range of software applications. Multimedia, 3D, and Internet application use has increased sharply over the past few years, and this trend is anticipated to continue in the future. For this reason, a wide range of benchmarks should be used to evaluate processor and system performance. PC users and buyers should consider the entire Spectrum of Performance, which includes productivity; compute intensive, Internet and entertainment performance. The Pentium 4 processor offers great performance for today's and tomorrow's applications developed for the Intel architecture, as well as the quality, reliability, and compatibility you have come to expect from the world's leading microprocessor company.

This performance brief introduces the Pentium 4 processor, explains the technologies that make it work, examines the purpose and methods behind the industry's most useful benchmarks, and shows how the Pentium 4 processor currently performs on each of the respective benchmarks. As new benchmarks are introduced, this performance brief will be updated as appropriate.

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INTRODUCTION

The Pentium® 4 processor is Intel's most advanced and powerful processor for desktop PCs and is designed to deliver performance across applications and usage models where PC users can truly appreciate and experience performance. The Pentium 4 processor delivers world-class performance for consumer enthusiasts, business professionals, as well as for entry-level workstation users. With all the performance needed to maximize both current and future Web technologies, as well as being designed to maximize investment protection, Pentium 4 processors will continue to deliver world-class performance for PC users well into the future.

When a Pentium 4 processor is used in conjunction with an Intel® 850 chipset family, the resulting platform provides the highest level of performance for today's software and performance headroom for tomorrow's emerging software and usage models. The Intel 850 chipset family implements the next step in the evolution of the Intel Hub Architecture and was designed in tandem with the Intel Pentium 4 processor and innovative Intel NetBurst micro-architecture. The Intel 850 chipset family was designed to maximize performance and enhance the PC user experience both on and off the Internet. Features such as 533 MHz and 400 MHz system bus offer a peak bandwidth of 4.2 GB/s and 3.2 GB/s, Dual RDRAM memory channels also offer a peak bandwidth of 4.2 GB/s or 3.2 GB/s that is balanced with the respective system bus, and the AGP 4X interface allows graphics controllers to access main memory at over 1 GB/s.

This brief provides performance results for the following Intel Pentium 4 processors while using an Intel 850 chipset family and running a variety of benchmarks:

- Pentium 4 processor 1.50 GHz
- Pentium 4 processor 1.60 GHz
- Pentium 4 processor 1.70 GHz
- Pentium 4 processor 1.80 GHz
- Pentium 4 processor 1.90 GHz
- Pentium 4 processor 2 GHz
- Pentium 4 processor 2A GHz
- Pentium 4 processor 2.20 GHz
- Pentium 4 processor 2.26 GHz
- Pentium 4 processor 2.40 GHz
- Pentium 4 processor 2.53 GHz
- Pentium 4 processor 2.66 GHz
- Pentium 4 processor 2.80 GHz

When evaluating the performance of a microprocessor or system, it is important to obtain the complete performance picture. Today's PC user runs a broad spectrum of applications consisting of productivity, computer intensive, Internet and entertainment:

- Productivity includes applications such as word processing, presentation, and personal finance programs.
- Computer intensive includes integer and floating point applications
- Internet applications include Internet browsers, as well as 3D and multimedia Web content.
- Entertainment includes audio, digital video, imaging, and gaming applications.

A processor and system should deliver the highest performance across the entire Spectrum of Performance: Productivity, Compute Intensive, Internet and Entertainment.

This report provides benchmark results for the Intel Pentium 4 processor family. Modern, industry-standard benchmarks were chosen to demonstrate capabilities across the Spectrum of Performance:

- Productivity performance can be measured using processor-level productivity benchmarks BAPCO's SYSmark* 2002 benchmark.
- Compute Intensive performance can be measured using system level component benchmarks such as SPEC* CPU2000.
- Internet performance can be measured by the BAPCO and MadOnion.com WebMark* 2001 benchmark.
- Entertainment performance can be determined by taking the performance on several metrics, which measure the performance of Audio, Digital Video Encoding, Imaging performance and Gaming performance.

Intel is committed to using the most robust and relevant benchmarks in characterizing the performance of its products, and will adapt this mix over time as newer benchmarks are introduced into the PC market.

System performance does not depend on the microprocessor alone. Hardware and software system components—such as the operating system, the graphics and I/0 subsystems, application software, and memory—may significantly affect benchmark results. For this reason, this performance brief illustrates Pentium 4 processor performance on a consistent system configuration. Details of the system configuration used for the benchmarks throughout this brief can be found in Appendix A.

SPECTRUM OF PERFORMANCE

When evaluating the performance of a microprocessor or system, it is important to obtain the complete performance picture. A processor and system should deliver high performance across the entire Spectrum of Performance: Productivity, Compute Intensive, Internet and Entertainment.

Figure 1. The Spectrum of Performance

Productivity Benchmarks

Productivity performance should represent current business usage models of multi-tasking with background computing. The Industry Standard productivity benchmark SYSmark* 2002 incorporates mainstream applications for office productivity and Internet content creation as well as the latest business usage models to reflect platform productivity performance.

- SYSmark* 2002

Compute Intensive Benchmarks

Compute intensive performance focuses on the component-level and should measure the performance of the computer's processor, chipset, memory architecture, and software compiler. The SPEC* CPU2000 benchmark suit is designed to provide component level performance on compute-intensive integer performance and compute-intensive floating-point performance.

- SPECfp*_base2000
- SPECint*_base2000
- NewTek* Lightwave 3D* 7.5

Internet Technology Benchmarks

Internet performance should represent the actual user experience a business/individual would experience while using the Internet. This should include scenarios for business-to-business, business-to-consumer as well as emulate the activities that occur on a corporate intranet. In addition each one of these scenarios should contain the technology that make up the backbone of the Internet such as Flash*, XML* and Java*. WebMark* 2001 is an Internet benchmark designed to measure the performance of a user interacting on the Internet.

- WebMark* 2001

Entertainment Benchmarks and Applications

Entertainment performance should encompass the activities that end users utilize a computer for. The list of activities include audio encoding using the MP3 file format and consists of converting a .wav file format to an .mp3 format. MAGIX* MP3 Maker* V.2.39 Gold is an application that lets a user create MP3 files from the .wav format. In addition to audio encoding, video encoding is an important aspect of entertainment performance. XMPEG* v.4.5 with DivX* 5.01 Codec takes MPEG streams and encodes them into DivX format so content can be easily shared or viewed. Imaging performance is also an important aspect of entertainment performance. Adobe* Photoshop* v.7.0 is an application that modifies images to be used for Internet content creation. Finally 3D performance, measured in frames per second, which represents significant demands on a computing platform, can be measured by online games as Quake* III Arena.

- Audio - MAGIX* MP3 Maker* V.2.39 Gold
- Digital Video - XMPEG* v.4.5 with DivX* 5.01 Codec
- Imaging - Adobe* Photoshop* 7.0
- Gaming - Quake* III Arena Demo Four

Evolving and Future Benchmarks

Benchmarks evolve as a direct response to the development of new technologies and the need to evaluate them objectively. As hardware and software technologies evolve, benchmarks must also evolve to capture new capabilities accurately.

The Pentium® 4 processor represents a major step in the advancement of computing performance. Currently available benchmarks show considerable gains across the Spectrum of Performance. As existing benchmarks are updated to support the new capabilities, and as new benchmarks are introduced, Intel will update this performance brief as appropriate.

THE INTEL® PENTIUM® 4 PROCESSOR

The Intel® Pentium® 4 processor, Intel's most advanced, most powerful processor, is based on the new Intel NetBurst™ micro-architecture. The Pentium 4 processor is designed to deliver performance across applications and usages where end users can truly appreciate and experience the performance. These applications include Internet audio and streaming video, image processing, video content creation, speech, 3D, CAD, games, multimedia, and multi-tasking user environments. The Intel Pentium 4 processor delivers this world-class performance for consumer enthusiast and business professional desktop users as well as for entry-level workstation users.

Highlights of the Pentium 4 processor:

- Available at speeds ranging from 1.50 to 2.80 GHz
- Featuring the new Intel NetBurst micro-architecture
- Supported by the Intel® 850 chipset family
- Fully compatible with existing Intel Architecture-based software
- Internet Streaming SIMD Extensions 2
- Intel® MMX™ media enhancement technology
- Memory cacheability up to 4 GB of addressable memory space and system memory scalability up to 64 GB of physical memory
- Support for uni-processor designs
- Based upon Intel's 0.13 micron and 0.18 micron manufacturing processes

INTEL® PENTIUM® 4 PROCESSOR PRODUCT FEATURE HIGHLIGHTS

The Intel NetBurst™ micro-architecture delivers a number of new and innovative features including Hyper Pipelined Technology, 533 MHz and 400 MHz System Bus (supported by the 850E and 850 chipset respectively), Execution Trace Cache, and Rapid Execution Engine as well as a number of enhanced features Advanced Transfer Cache, Advanced Dynamic Execution, Enhanced Floating-point and Multimedia Unit, and Streaming SIMD Extensions 2. Many of these new innovations and advances were made possible with improvements in processor technology, process technology, and circuit design that could not previously be implemented in high-volume, manufacturable solutions. The features and resulting benefits of the new micro-architecture are defined below.

- Hyper Pipelined Technology:
  The hyper-pipelined technology of the NetBurst micro-architecture doubles the pipeline depth compared to the P6 micro-architecture used on today's Pentium III processors. One of the key pipelines, the branch prediction / recovery pipeline, is implemented in 20 stages in the NetBurst micro-architecture, compared to 10 stages in the P6 micro-architecture. This technology significantly increases the performance, frequency, and scalability of the processor.

- 533 MHz and 400 MHz System Bus:
  The Pentium 4 processor supports Intel's highest performance desktop system bus by delivering 4.2 GB and 3.2 GB, of data per second into and out of the processor. This is accomplished through a physical signaling scheme of quad pumping the data transfers over a 100-MHz clocked system bus and a buffering scheme allowing for sustained 400-MHz data transfers. This compares to 1.06 GB/s delivered on the Pentium III processor's 133-MHz system bus.

- Level 1 Execution Trace Cache:
  In addition to the 8KB data cache, the Pentium 4 processor includes an Execution Trace Cache that stores up to 12K decoded micro-ops in the order of program execution. This increases performance by removing the decoder from the main execution loop and makes more efficient usage of the cache storage space since instructions that are branched around are not stored. The result is a means to deliver a high volume of instructions to the processor's execution units and a reduction in the overall time required to recover from branches that have been mis-predicted.

- Rapid Execution Engine:
  Two Arithmetic Logic Units (ALUs) on the Pentium 4 processor are clocked at twice the core processor frequency. This allows basic integer instructions such as Add, Subtract, Logical AND, Logical OR, etc. to execute in half a clock cycle. For example, the Rapid Execution Engine on a 1.50 GHz Pentium 4 processor runs at 3 GHz.

- 256KB or 512KB, Level 2 Advanced Transfer Cache:
  The Level 2 Advanced Transfer Cache (ATC) is either 256KB or 512KB in size and delivers a much higher data throughput channel between the Level 2 cache and the processor core. The Advanced Transfer Cache consists of a 256-bit (32-byte) interface that transfers data on each core clock. As a result, the Pentium 4 processor 1.50 GHz can deliver a data transfer rate of 48 GB/s. This compares to a transfer rate of 16 GB/s on the Pentium III processor at 1 GHz. Features of the ATC include:

  ⇒ Non-Blocking, full speed, on-die Level 2 cache
  ⇒ 8-way set associative
  ⇒ 256-bit data bus to the level 2 cache
  ⇒ Data clocked into and out of the cache every clock cycle
  ⇒ 256KB Level 2 ATC found on Pentium 4 processors at 1.30 GHz, 1.40 GHz, 1.50 GHz, 1.60 GHz, 1.70 GHz, 1.80 GHz, 1.90 GHz, and 2 GHz
  ⇒ 512KB Level 2 ATC found on Pentium 4 processors at 2A GHz, 2.20 GHz, 2.26 GHz, 2.40 GHz, 2.53 GHz, 2.66 GHz and 2.80 GHz

- Advanced Dynamic Execution:
  The Advanced Dynamic Execution engine is a very deep, out-of-order speculative execution engine that keeps the execution units executing instructions. The Pentium 4 processor can also view 126 instructions in flight and handle up to 48 loads and 24 stores in the pipeline. It also includes an enhanced branch prediction algorithm that has the net effect of reducing the number of branch mis-predictions by about 33% over the P6 generation processor's branch prediction capability. It does this by implementing a 4KB branch target buffer that stores more detail on the history of past branches, as well as by implementing a more advanced branch prediction algorithm.

- Enhanced Floating-Point and Multimedia Unit:
  The Pentium 4 processor expands the floating-point registers to a full 128-bit and adds an additional register for data movement which improves performance on both floating-point and multimedia applications.

- Internet Streaming SIMD Extensions 2 (SSE2):
  With the introduction of SSE2, the NetBurst™ micro-architecture now extends the SIMD capabilities that MMX™ technology and SSE technology delivered by adding 144 new instructions. These instructions include 128-bit SIMD integer arithmetic and 128-bit SIMD double-precision floating-point operations. These new instructions reduce the overall number of instructions required to execute a particular program task and as a result can contribute to an overall performance increase. They accelerate a broad range of applications, including video, speech, and image, photo processing, encryption, financial, engineering and scientific applications.

- Features Used for Test and Performance / Thermal Monitoring:
  ⇒ Built-in Self Test (BIST) provides single stuck-at fault coverage of the microcode and large logic arrays, as well as testing of the instruction cache, data cache, Translation Lookaside Buffers (TLBs), and ROMs.
  ⇒ IEEE 1149.1 Standard Test Access Port and Boundary Scan mechanism enables testing of the Pentium 4 processor and system connections through a standard interface.
  ⇒ Internal performance counters can be used for performance monitoring and event counting.

⇒ Includes a new Thermal Monitor feature that allows motherboards to be cost effectively designed to expected application power usages rather than theoretical maximums.

INTEL® 850 CHIPSET FAMILY PRODUCT FEATURE HIGHLIGHTS

As the next step in the evolution of the Intel® Hub Architecture, the Intel® 850 chipset family was designed in tandem with the Intel® Pentium® 4 processor and innovative Intel® NetBurst™ micro-architecture.

Intel Hub Architecture delivers twice the I/O bandwidth as previous generation northbridge / southbridge technology. With dedicated data paths to fully optimize the additional bandwidth, the Intel 850 chipset family offers significant gains in the performance PC market and support for future Intel NetBurst micro-architecture based processors.

The 82850 Memory Controller Hub (MCH) delivers dual RDRAM memory channels and a 533 MHz and 400 MHz system bus, providing the latest graphics support through 1.5V AGP4X technology. Together these features deliver the highest total bandwidth capabilities to the PC platform.

The enhanced 82801BA I/O Controller Hub (ICH2) delivers twice the I/O bandwidth over traditional bridge architecture and provides dedicated data paths to fully optimize the additional bandwidth. The ICH2 makes a direct connection from the graphics and memory for faster access to peripherals and provides the features and bandwidth required for the performance PC.

In addition to advanced application support, the Intel 850 chipset family was designed with the following feature to enhance the end-user experience both on and off the Internet:

- 533 MHz and 400 MHz system bus delivers a high bandwidth connection between the Intel Pentium 4 processor and the platform. To ensure maximum performance, the system bus is balanced with the dual RDRAM channels at 4.2 GB/s and 3.2 GB/s corresponding to the respected system bus, up to 4x the bandwidth of platforms based on Intel® Pentium® III processors.

- Intel Hub Architecture with optional P64H provides increased I/O bus bandwidth to allow better concurrency for media-rich applications and multi-tasking

- Dual RDRAM memory channels deliver 4.2 GB/s or 3.2 GB/s of memory bandwidth to the processor with respective system bus. High memory bandwidth, coupled with an efficient protocol, deliver balanced platform support and provide the memory bandwidth necessary to extract full performance from the Intel Pentium 4 processor.

- The AGP4X interface allows graphics controllers to access main memory at over 1 GB/s, twice that of previous AGP platforms. With RDRAM memory and the new Intel Pentium 4 processor, AGP4X delivers the next level of 3D graphics performance

- Two USB controllers double the bandwidth available for USB peripherals to 24 MB/s over four ports. This results in a significant increase over previous integrated 1-4 port hubs at 12 MB/s.

- The latest AC97 audio delivers six channels of audio for enhanced sound quality and full surround sound capability for live broadcast and other Digital Dashboard programming.

- LAN Connect Interface (LCI) provides flexible network solutions such as home phone line, 10/100 Mbps Ethernet, and 10/100 Mbps Ethernet with LAN manageability.

- Dual Ultra ATA-100 controllers support the fastest IDE interface for transfers to storage devices. Additional performance is gained with Intel's Storage Driver over standard ATA drivers.

- Communication and Network Riser (CNR) offers flexibility in system configuration with a baseline feature set that can be upgraded with an audio card, modem card, or network card.

Designed to balance the performance and power in the new Intel® Pentium® 4 processor, the Intel® 850 chipset family delivers a robust foundation for the most sophisticated end-user applications. The 850 chipset family offers innovative design, high-speed memory, and configuration options that optimized performance and provide a solid base for the Intel Pentium 4 processor with the Intel® NetBurst™ micro-architecture.

PERFORMANCE SUMMARY

When evaluating the performance of a microprocessor or system, it is important to obtain the complete performance picture. A processor and system should deliver high performance across the entire Spectrum of Performance: Productivity, Compute Intensive, Internet, and Entertainment.

Productivity Performance

Productivity performance should represent current business usage models of multi-tasking with background computing. The Industry Standard productivity benchmark SYSmark* 2002 incorporates mainstream applications for office productivity and Internet content creation as well as the latest business usage models to reflect platform productivity performance.

SYSmark* 2002

SYSmark* 2002 is a suite of application software and associated benchmark workloads developed by the Business Applications Performance Corporation (BAPCO), a non-profit consortium of leading computer industry publications, independent testing labs, PC hardware manufacturers, semiconductor manufacturers, and software publishers. SYSmark 2002 is a tool that measures system performance on popular business-oriented applications in the Microsoft* Windows* operating environment.

SYSmark 2002 contains fourteen application workloads that are divided into two categories:

Office Productivity:
- Dragon* Naturally Speaking* Preferred Version 5
- McAfee* VirusScan* 5.13
- Microsoft Access* 2002
- Microsoft Excel* 2002
- Microsoft Outlook* 2002
- Microsoft PowerPoint* 2002
- Microsoft Word* 2002
- Netscape* Communicator* 6.0
- WinZip* 8.0

Internet Content Creation:
- Adobe* Photoshop* 6.0.1
- Adobe Premiere* 6.0
- Macromedia* Dreamweaver* 4
- Macromedia Flash* 5
- Microsoft Windows Media Encoder* 7.1

Figure 2 illustrates the SYSmark 2002 rating under Windows* XP for the Pentium® 4 processor.

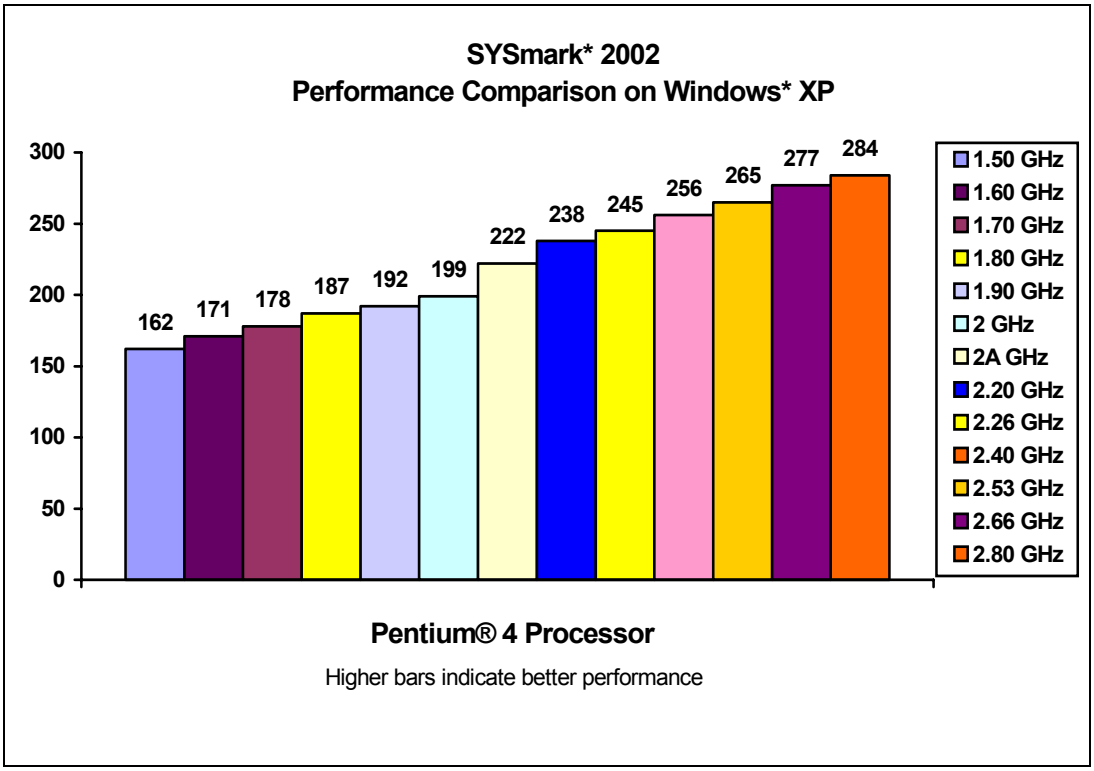

Figure 2. Intel® Pentium® 4 Processor Performance for SYSmark* 2002 on Windows* XP

Compute Intensive Performance

SPEC* CPU2000

Compute intensive performance focuses on the component-level and should measure the performance of the computer's processor, chipset, memory architecture, and software compiler. The SPEC* CPU2000 benchmark suit is designed to provide component level performance on compute-intensive integer performance and compute-intensive floating-point performance.

SPEC* CPU2000 is a software benchmark product that can be run on Windows* NT, Windows* 2000, and many varieties of UNIX*. SPEC* CPU2000 is produced by the Standard Performance Evaluation Corp. (SPEC), a non-profit group of computer vendors, system integrators, universities, research organizations, publishers, and consultants throughout the world. It was designed to provide comparisons of performance for compute-intensive workloads on different computer systems.

SPEC* CPU2000 consists of two suites of benchmarks: SPECint*2000 for measuring and comparing compute-intensive integer performance, and SPECfp*2000 for measuring and comparing compute-intensive floating-point performance. The two suites provide component-level benchmarks that measure the performance of the computer's processor, memory architecture, and compiler. SPEC benchmarks are selected from existing application and benchmark source code running across multiple platforms.

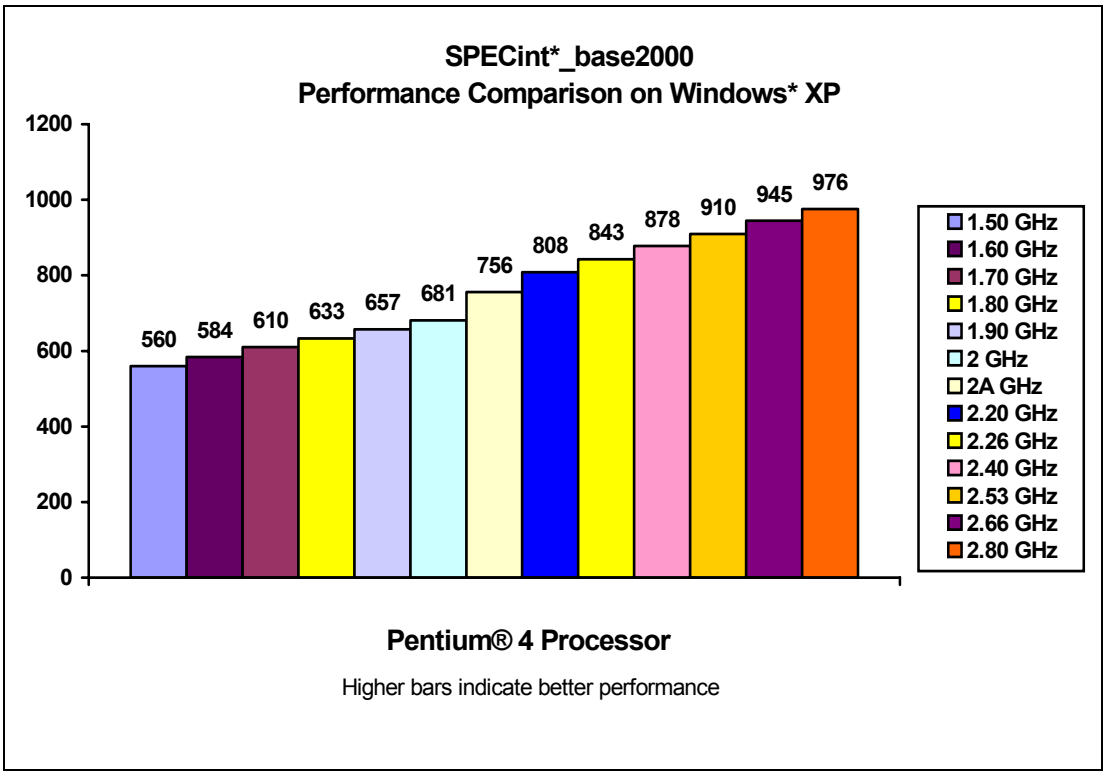

Figure 3. Intel® Pentium® 4 Processor Performance for the SPECint*_base2000 Benchmark

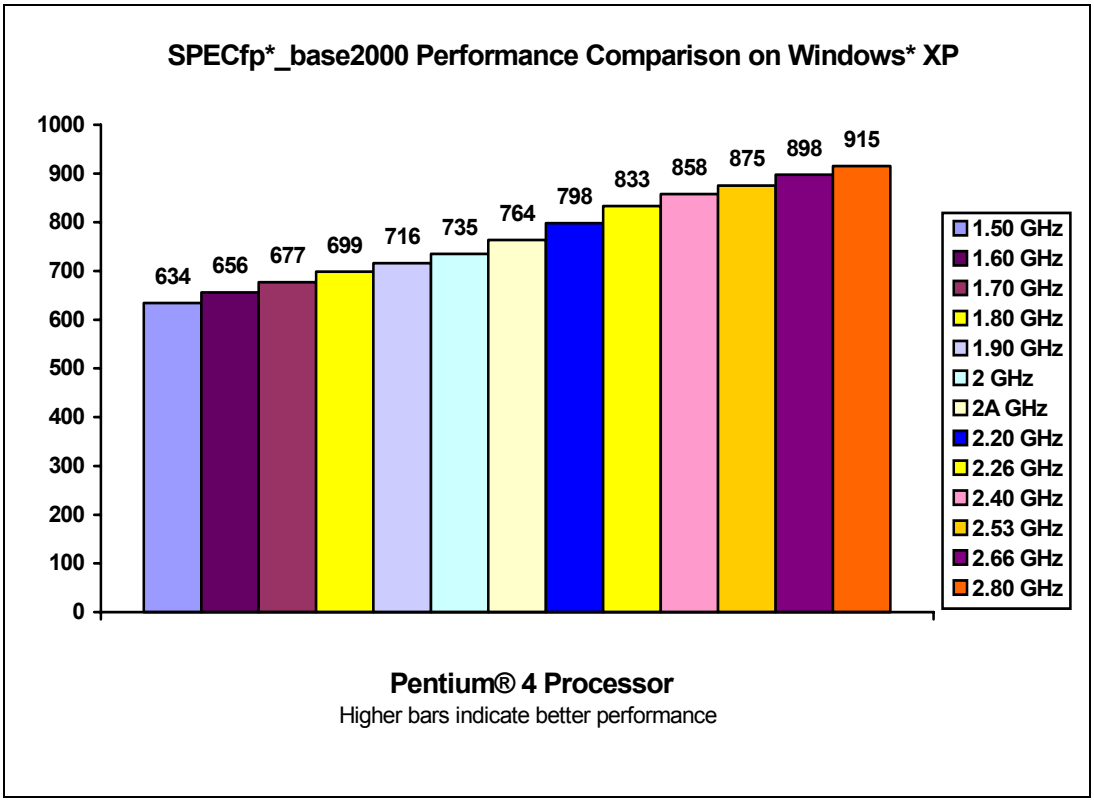

Figure 4. Intel® Pentium® 4 Processor Performance for the SPECfp*_base2000 Benchmark

NewTek* LightWave 3D*

Rendering applications demonstrate the new levels of complexity that can be achieved by running the latest software on powerful systems. With these tools new content can be created to enhance 3D solutions. The NewTek LightWave 3D* 7.5 software evaluation guide highlights the compute intensive 3D rendering performance.

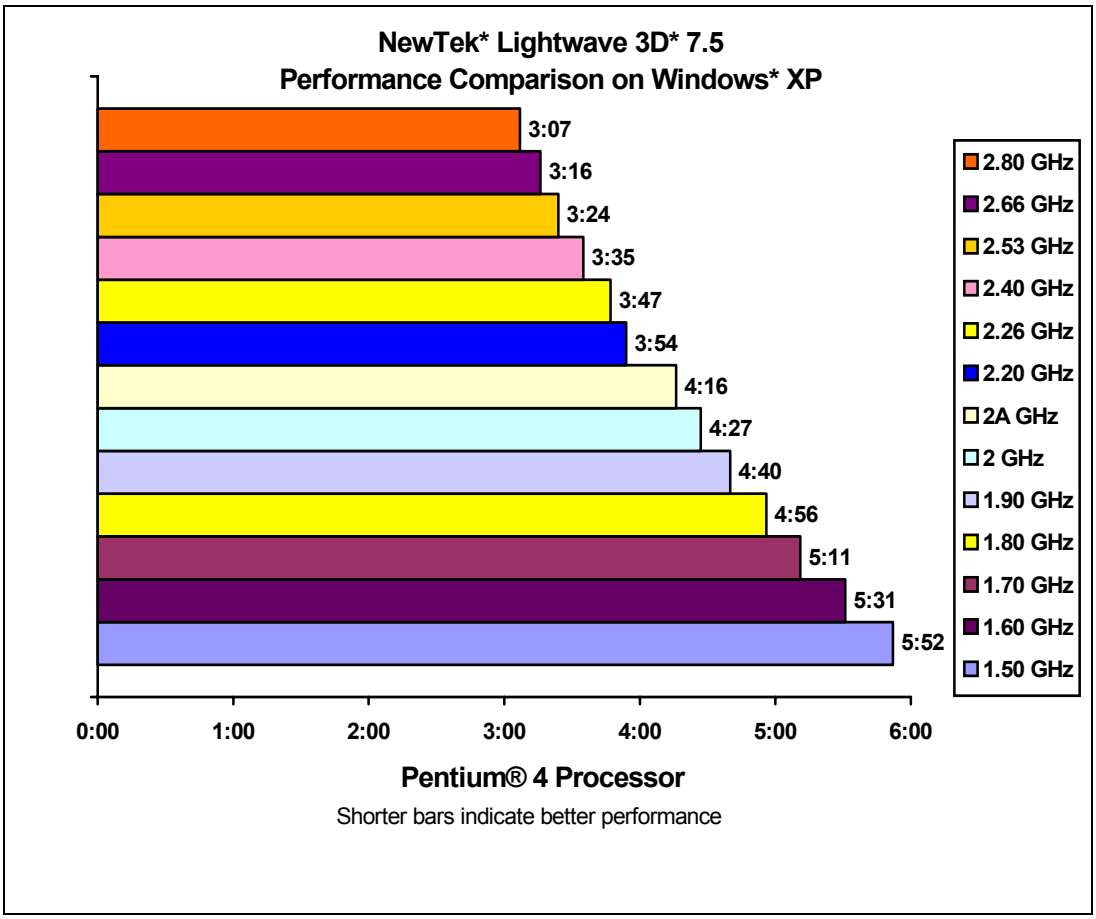

Figure 5. Intel® Pentium® 4 Processor Performance for the NewTek* LightWave 3D* 7.5 Software Evaluation Guide on Windows*XP

Internet Technology Performance

WebMark* 2001

Internet performance should represent the actual user experience a business/individual would experience while using the Internet. This should include scenarios for business-to-business, business-to-consumer as well as emulate the activities that occur on a corporate intranet. In addition each one of these scenarios should contain the technology that make up the backbone of the Internet such as Flash*, XML* and Java*. WebMark* 2001 is an Internet benchmark designed to measure the performance of a user interacting on the Internet.

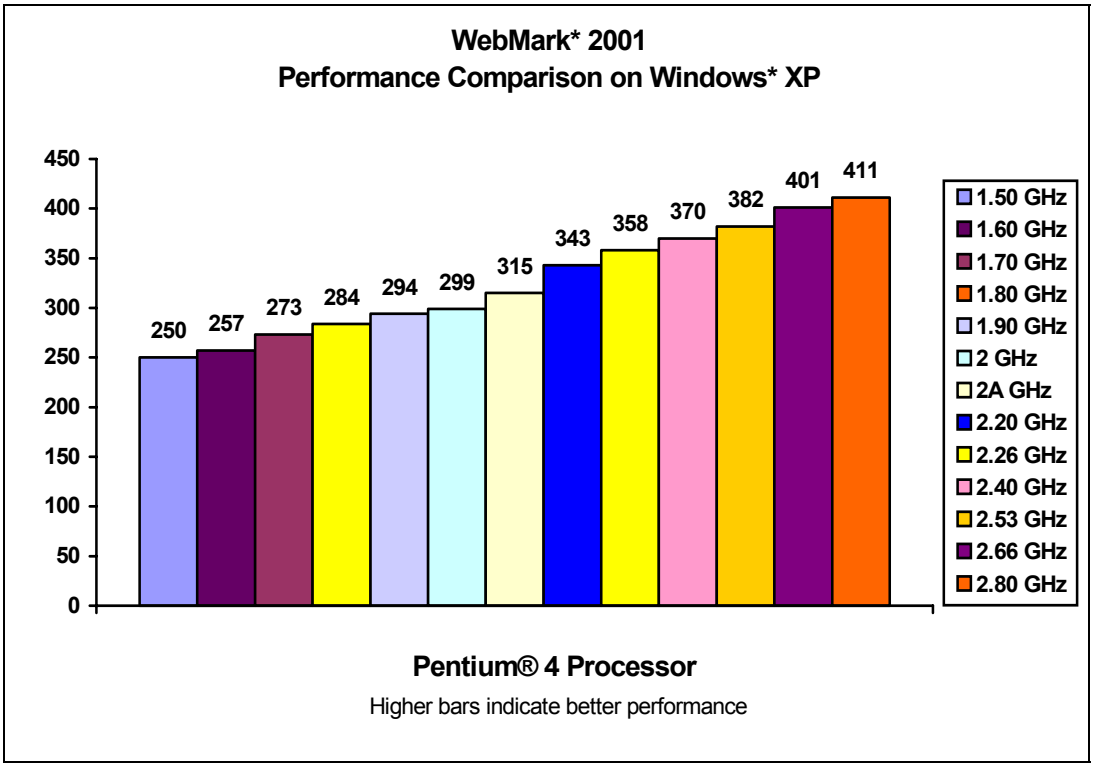

Figure 6. Intel® Pentium® 4 Processor Performance for WebMark* 2001 on Windows* XP

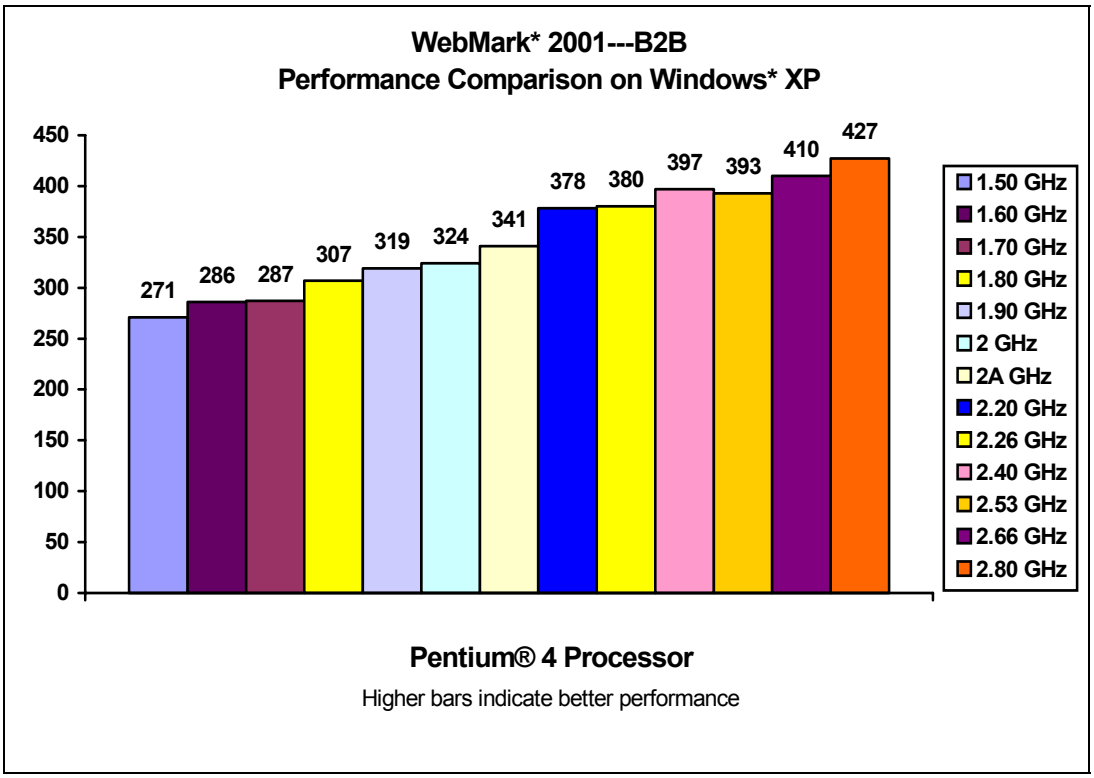

Figure 7. Intel® Pentium® 4 Processor Performance for WebMark* 2001—B2B on Windows* XP

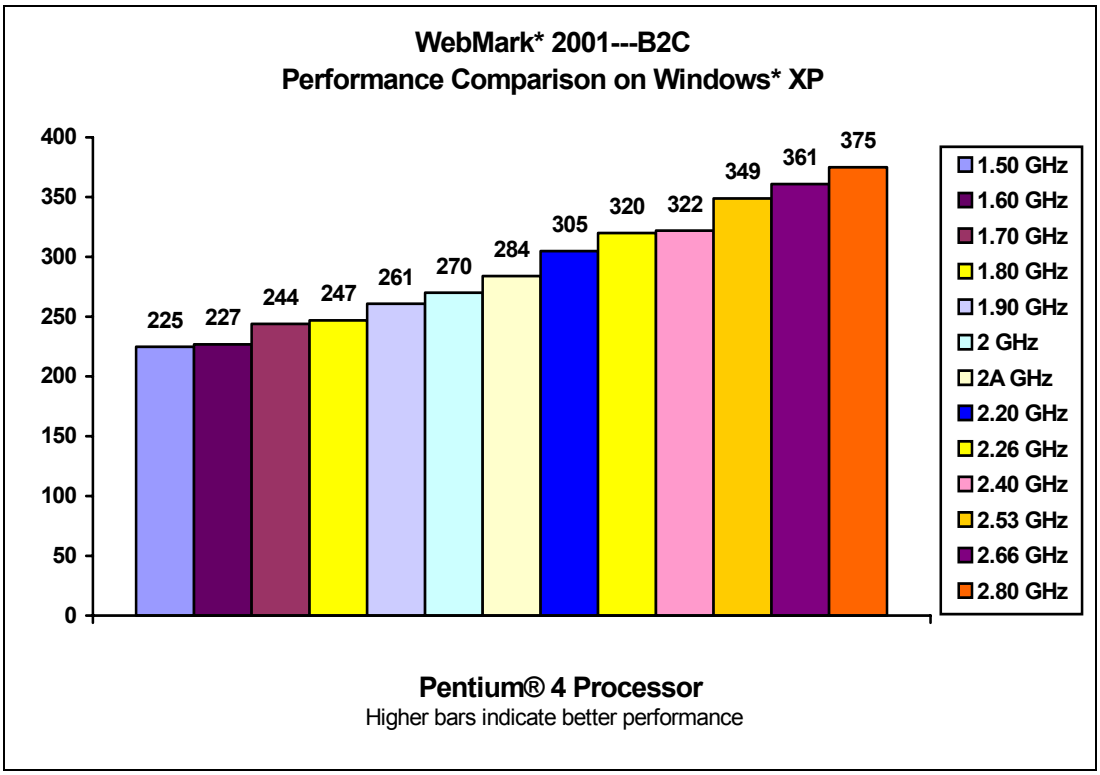

Figure 8. Intel® Pentium® 4 Processor Performance for WebMark* 2001—B2C on Windows* XP

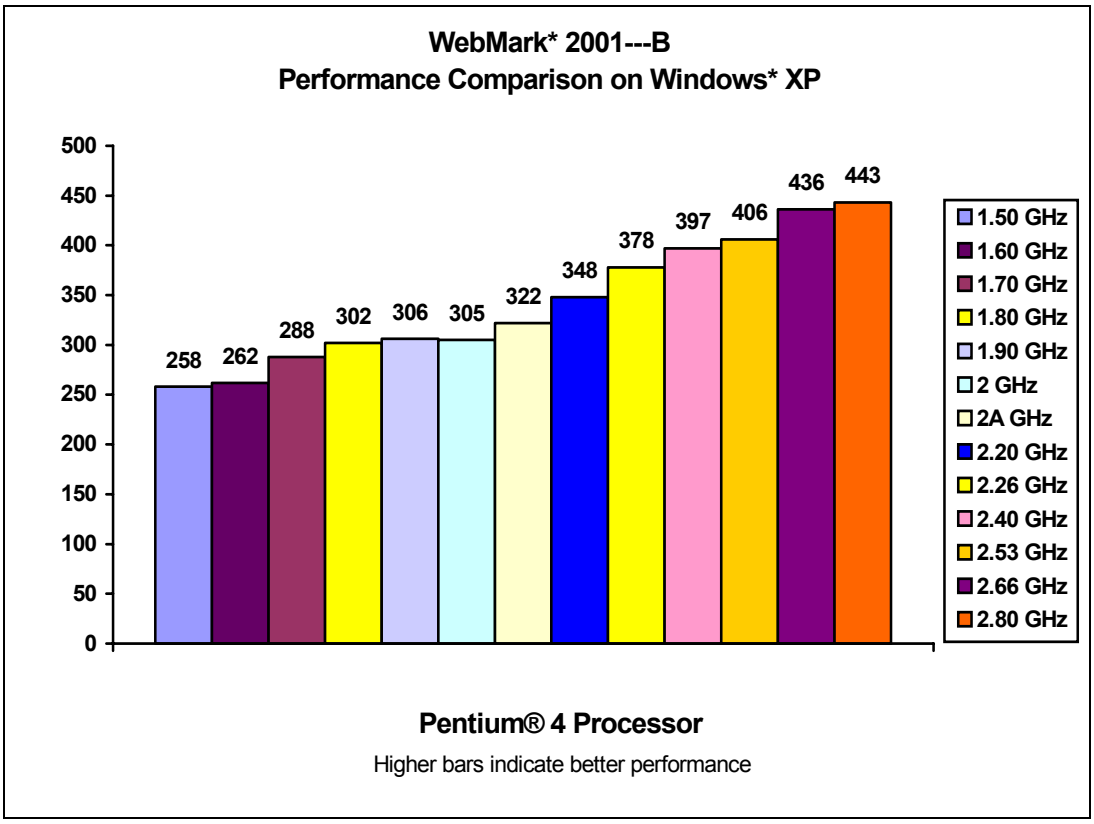

Figure 9. Intel® Pentium® 4 Processor Performance for WebMark* 2001—B on Windows* XP

Entertainment Performance

Entertainment performance should encompass the activities that end users utilize a computer for. This should include audio encoding using the MP3 file format and consist of converting a .wav file format to an .mp3 format. MAGIX* MP3 Maker* V.2.39 Gold is an application that lets a user create MP3 files from the .wav format. In addition to audio encoding, video encoding is an important aspect of entertainment performance. XMPEG* v.4.5 with DivX* 5.01 Codec takes MPEG streams and encodes them into another video format so content can be easily shared or viewed. Imaging performance is also an important aspect of entertainment performance. Adobe* Photoshop* 7.0 is an application that modifies images to be used for Internet content creation. Finally 3D performance, measured in frames per second, which represents significant demands on a computing platform, can be measured by online games as Quake* III Arena.

Audio - MP3 Encoding

Sharing music is one of many popular uses of the Internet today. The MP3 file format is one of the more commonly used file formats used to share music since it enables high-fidelity sound with a relatively small file size. One of the more compute intensive steps in creating an MP3 file involves an encoding process going from a .wav file format to a .mp3 file format and there are software applications available today that help make this process easy. MAGIX* MP3 Maker* V.2.39 Gold is an application that lets a user create MP3 files from an audio CD.

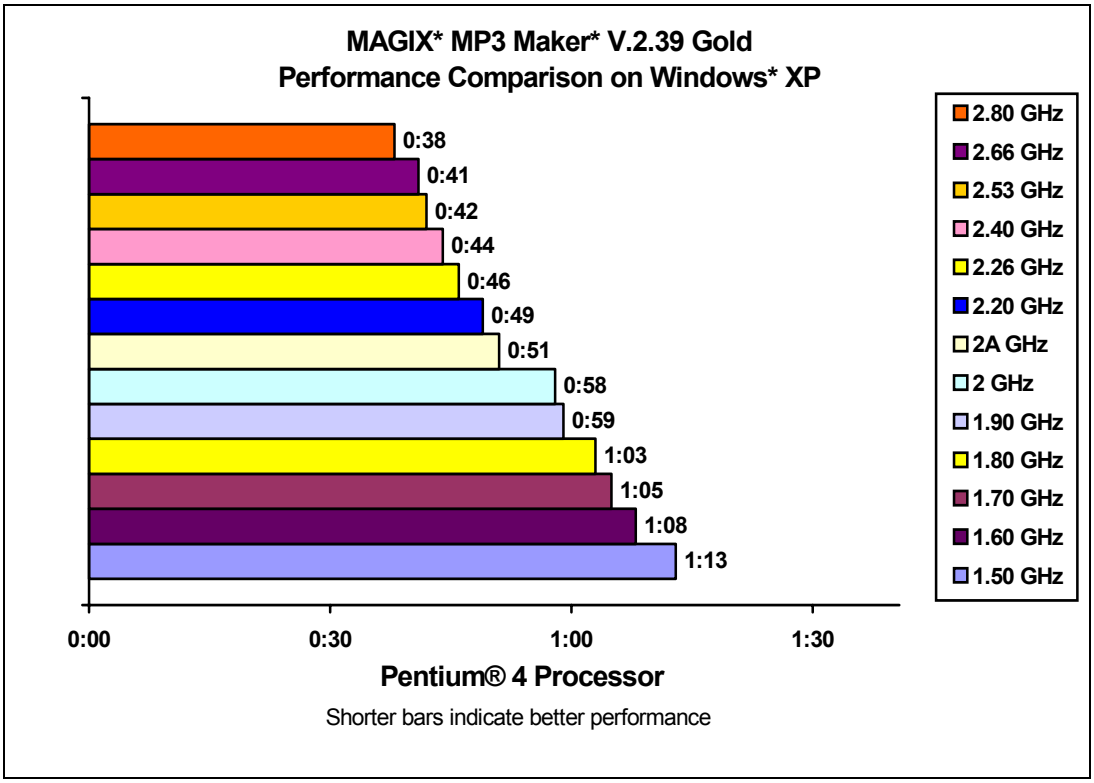

Figure 10. Intel® Pentium® 4 Processor Performance for the MAGIX* MP3 Maker* V.2.39 Gold Benchmark

Digital Video Encoding

Often before digital video is shared over the Internet it is either encoded into a file format that supports compression of the digital video into a smaller file size, or it is encoded into one of the popular video streaming formats commonly found on the Internet. XMPEG* v.4.5 with DivX* 5.01 Codec takes MPEG streams and encodes them into another video format. Maximum quality is achieved while minimizing the conversion impact and provides a straightforward approach to the MPEG 1&2 to MPEG 4 or AVI conversion.

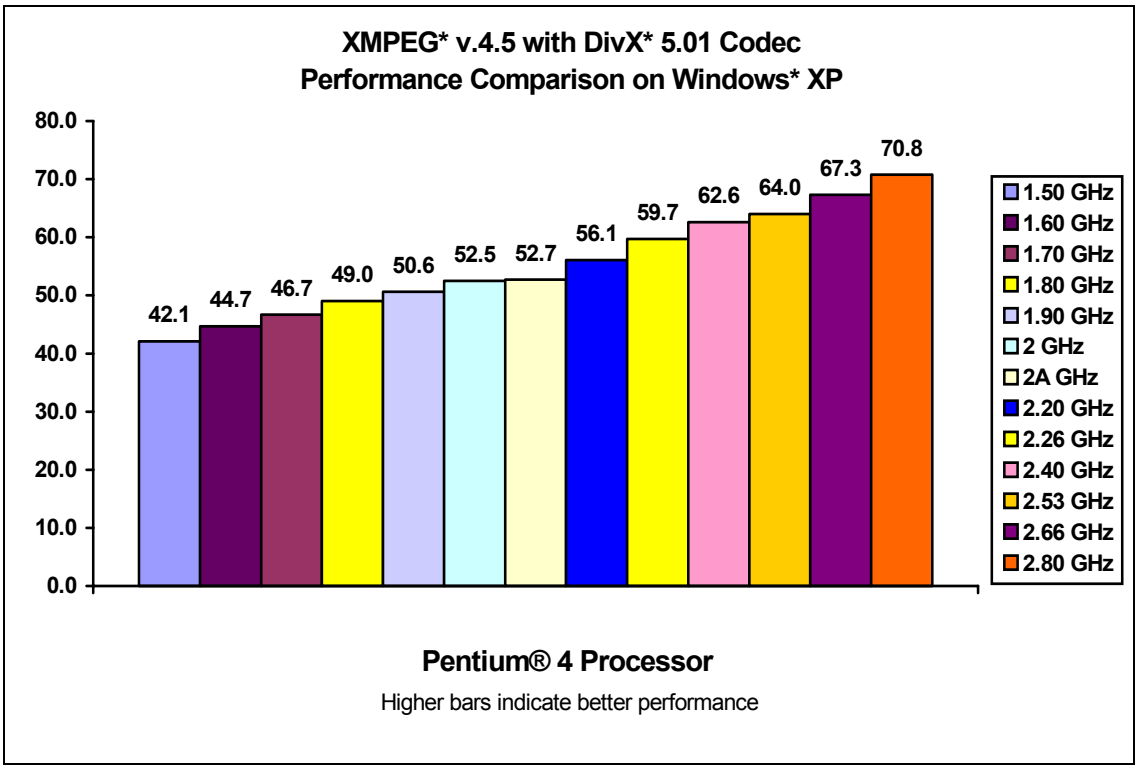

Figure 11. Intel® Pentium® 4 Processor Performance for the XMPEG* v.4.5 with DivX* 5.01 Codec Software Evaluation Guide on Windows* XP

Imaging Performance

Imaging performance is also an important aspect of entertainment performance. Adobe* Photoshop* 7.0 is an application that modifies images to be used for Internet content creation.

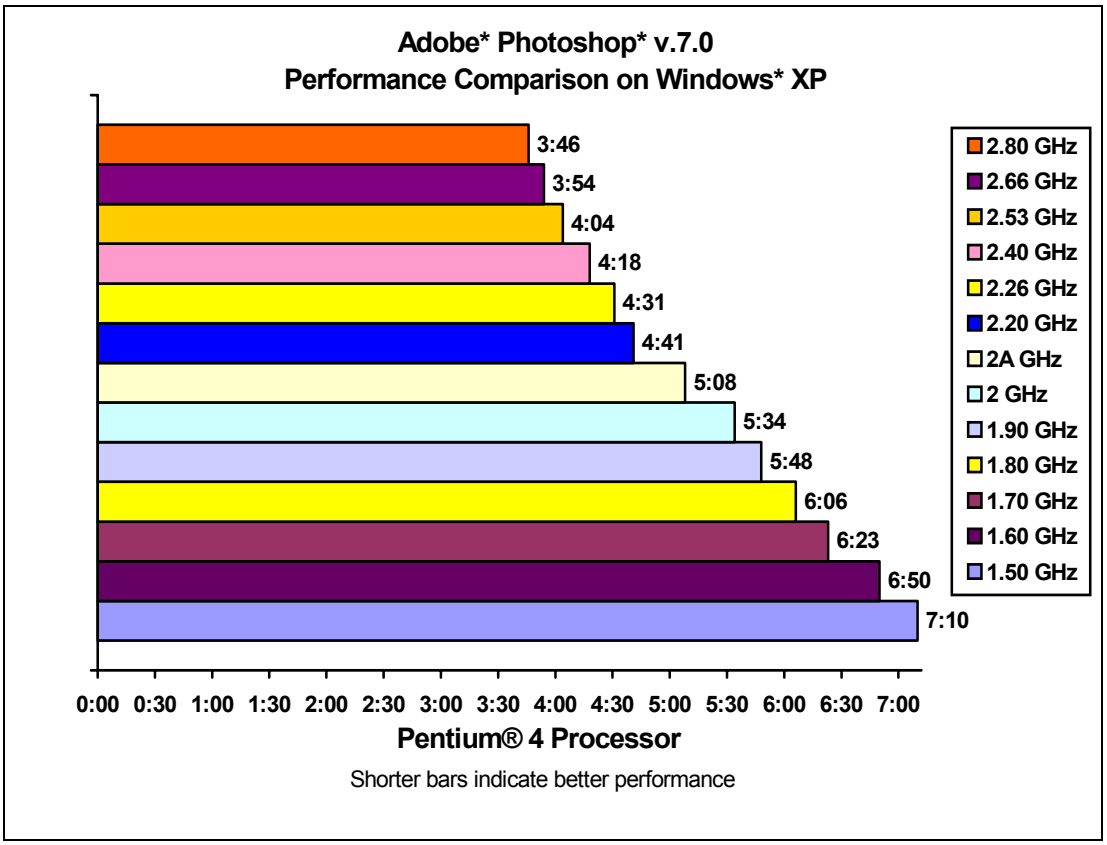

Figure 12. Intel® Pentium® 4 Processor Performance for Adobe* Photoshop* 7.0 Software Evaluation Guide on Windows*XP

Gaming Performance

Gamers tend to be very discerning when it comes to high-quality, high-performance 3D graphics and they often demand the best in terms of technology. One of the various metrics used to evaluate 3D performance is frames per second. Higher frames per second indicates smoother game play. The increasing compute-intensive demands of 3D games make it critical to continue to offer great gains for this type of software. Quake* III Arena is one of the hottest online games on the Internet today and includes one test that gamers use to evaluate 3D gaming performance- Quake III Arena Demo Four.

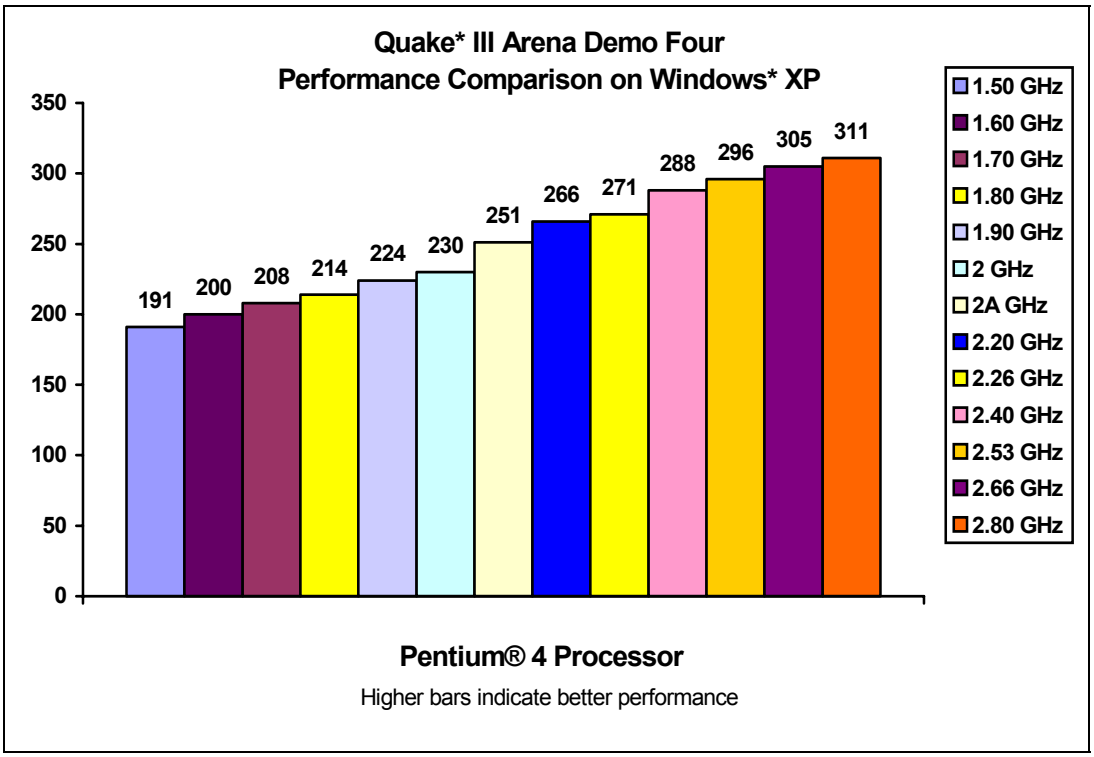

Figure 13. Intel® Pentium® 4 Processor Performance for the Quake* III Arena Demo Four Benchmark

SUMMARY

Table 1 summarizes productivity benchmark performance for the Intel® Pentium® 4 processor. For time related scores less time indicates better performance. For all others a higher score indicates better performance.

Table 1. Spectrum of Performance Benchmark Results—Productivity Benchmarks

Pentium®4 processor	1.50 GHz	1.60 GHz	1.70 GHz	1.80 GHz	1.90 GHz	2 GHz	2A GHz[1]	2.20 GHz[1]	2.26 GHz[1,2]	2.40 GHz[1,2]	2.53 GHz[1,2]	2.66 GHz[1,2]	2.80 GHz[1,2]
SYSmark* 2002	162	171	178	187	192	199	222	238	245	256	265	277	284

[1]Intel's Advanced 0.13 micron process and 512K L2 Cache

[2]533 MHz Front Side Bus

Table 2 summarizes compute intensive benchmark performance for the Intel Pentium 4 processor. For time related scores less time indicates better performance. For all others a higher score indicates better performance.

Table 2. Spectrum of Performance Benchmark Results—Compute Intensive Benchmarks

Pentium®4 processor	1.50 GHz	1.60 GHz	1.70 GHz	1.80 GHz	1.90 GHz	2 GHz	2A GHz[1]	2.20 GHz[1]	2.26 GHz[1, 2]	2.40 GHz[1,2]	2.53 GHz[1,2]	2.66 GHz[1,2]	2.80 GHz[1,2]
SPECint*_base2000	560	584	610	633	657	681	756	808	843	878	910	945	976
SPECint*2000	562	588	613	638	660	684	759	811	847	882	914	952	984
SPECfp*_base2000	634	656	677	699	716	735	764	798	833	858	875	898	915
SPECfp*2000	643	666	687	708	726	745	773	808	846	870	889	911	929
NewTek* LightWave 3D* 7.5	5 min 52 sec	5 min 31 sec	5 min 11 sec	4 min 56 sec	4 min 40 sec	4 min 27 sec	4 min 16 sec	3 min 54 sec	3 min 47 sec	3 min 35 sec	3 min 24 sec	3 min 16 sec	3 min 07 sec

[1]Intel's Advanced 0.13 micron process and 512K L2 Cache

[2]533 MHz Front Side Bus

Table 3 summarizes Internet benchmark performance for the Intel Pentium 4 processor. A higher score indicates better performance.

Table 3. Spectrum of Performance Benchmark Results—Internet Technology Benchmarks

Pentium® 4 processor	1.50 GHz	1.60 GHz	1.70 GHz	1.80 GHz	1.90 GHz	2 GHz	2A GHz[1]	2.20 GHz[1]	2.26 GHz[1,2]	2.40 GHz[1,2]	2.53 GHz[1,2]	2.66 GHz[1,2]	2.80 GHz[1,2]
WebMark* 2001	250	257	273	284	294	299	315	343	358	370	382	401	411
WebMark* 2001–B2B	271	286	287	307	319	324	341	378	380	397	393	410	427
WebMark* 2001–B2C	225	227	244	247	261	270	284	305	320	322	349	361	375
WebMark* 2001–B	258	262	288	302	306	305	322	348	378	397	406	436	443

[1]Intel's Advanced 0.13 micron process and 512K L2 Cache

[2]533 MHz Front Side Bus

Table 4 summarizes entertainment benchmark performance for the Intel® Pentium® 4 processor. For time related scores less time indicates better performance. For all others a higher score indicates better performance.

Table 4. Spectrum of Performance Benchmark Results—Entertainment Benchmarks

Pentium®4 processor	1.50 GHz	1.60 GHz	1.70 GHz	1.80 GHz	1.90 GHz	2 GHZ	2A GHz[1]	2.20 GHz[1]	2.26 GHz[1,2]	2.40 GHz[1,2]	2.53 GHz[1,2]	2.66 GHz[1,2]	2.80 GHz[1,2]
MAGIX* MP3 Maker* V.2.39 Gold	1 min 13 sec	1 min 08 sec	1 min 05 sec	1 min 03 sec	0 min 59 sec	0 min 58 sec	0 min 51 sec	0 min 49 sec	0 min 46 sec	0 min 44 sec	0 min 42 sec	0 min 41 sec	0 min 38 sec
XMPEG* v.4.5 with DivX* 5.011 Codec	42.1	44.7	46.7	49.0	50.6	52.5	52.7	56.1	59.7	62.6	64.0	67.3	70.8
Adobe* Photoshop* v.7.0	7 min 10 sec	6 min 50 sec	6 min 23 sec	6 min 06 sec	5 min 48 sec	5 min 34 sec	5 min 08 sec	4 min 41 sec	4 min 31 sec	4 min 18 sec	4 min 04 sec	3 min 54 sec	3 min 46 sec
Quake* III Arena Demo Four	191	200	208	214	224	230	251	266	271	288	296	305	311

[1]Intel's Advanced 0.13 micron process and 512K L2 Cache

[2]533 MHz Front Side Bus

APPENDIX A — TEST CONFIGURATIONS

Table 5. System Configuration Used in Benchmark Tests

Processor	**Pentium® 4 Processor 1.50, 1.60, 1.70, 1.80, 1.90, 2, 2A[1], 2.20[1], 2.26[1,2], 2.40[1,2], 2.53[1, 2], 2.66[1, 2, 3], 2.80 GHz[1, 2, 3]**
Motherboard	Intel® Desktop Board D850EMV2
Motherboard BIOS	MV85010A.86A.0025.P10 [3] MV85010A.86A.0036.P14
FPU	Integrated
Secondary Cache	256KB or 512KB full-speed Advanced Transfer Cache
Memory Size	256 MB PC800-40 RDRAM (Samsung PC800-40 MR16R1624AF0-CM8_0214)
Hard Disk Controller/Bus	Intel® Application Accelerator v2.2.2128 Intel® Chipset Software Installation Utility v4.00.1009
Hard Disk	IBM* 120GXP 80 GB IC35L080AVVA07-0 ATA-100
Video Controller/Bus	Leadtek* WinFast* A250 Ultra TD nVIDIA* GeForce* 4 TI 4600 4x AGP graphics
Video Memory Size/Type	128MB DDRRAM
Operating System	Windows* XP w/ DirectX 8.1
Operating System settings	Visual Effects OFF, System Restore OFF, Windows Update OFF, Power Management OFF
Video Driver Revision	nVIDIA* Detonator* 4 reference driver 28.32
Graphics	1024x768 resolution, 32-bit color
CD ROM Drive	Toshiba* 32X XM-6302B IDE
Sound Card	Creative Labs SoundBlaster* Audigy PCI (Gamer version. Model # SB0090)
Network Card	Intel Pro/100 S desktop PCI LAN card. Model # 751767
SPEC* CPU2000 Compiler	Intel C/C++* and FORTRAN Compiler Plug-in V5.01 Microsoft Visual C/C++ 6.0 (for libraries)

[1]Intel's Advanced 0.13 micron process and 512K L2 Cache, [2] 533 MHz Front Side Bus

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